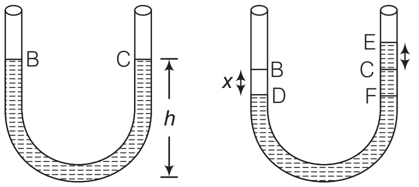
- **Frequency** is the number of full oscillations completed by pendulum in one second.
- Second pendulum is one whose time period is 2s.
- Time period of simple pendulum $T = 2\pi\sqrt{\frac{L}{g}}$, where L is length of thread and g is the acceleration due to gravity.
- The time period of a simple pendulum of infinite length is 84.6 min. A pendulum clock goes slow in summer and fast in winter.
- If a simple pendulum is suspended in a lift descending down with acceleration, then time period of pendulum will increase. If lift is ascending, then time period of pendulum will decrease.
- If a lift falling freely under gravity, then the time period of the pendulum will be infinite.
- At moon, the time period of simple pendulum increases because acceleration due to gravity at moon is lesser.

➠ **Note** Every body inside a satellite remains in a state of weightlessness, that is the effective value of g remains zero. Hence, a pendulum inside a satellite will not oscillate. Therefore, the pendulum clock cannot work inside space satellites.

Oscillation of Liquid Column in a U-tube

If a liquid is taken in a U-tube, it maintains equal level in both the columns of the tube. When air is blown-in on side B, the liquid may go down upto point D. The displacement of the liquid is given by $BD = x$.



In the second column, the liquid goes upto E and the displacement is CE. As the applied force is withdrawn, the liquid level in the second column comes down because of gravity from E to C. But due to inertia, it overshoots the mark C and goes to F ($CF = x$), whereas the liquid in the other column also goes up by the same distance.

Thus, the liquid starts oscillating in both the columns. This motion of the liquid in the U-tube is SHM with time period, $T = 2\pi\sqrt{\frac{h}{g}}$ where, h is the height of liquid in each column.

WAVES

A wave is a disturbance which propagates energy from one place to the other without the transport of matter.

These are of two types

Mechanical Waves

The waves which require material medium (solid, liquid or gas) for their propagation are called mechanical waves or elastic waves. These are of two types

(*i*) **Longitudinal Waves** If the particles of the medium vibrate in the direction of propagation of wave, the wave is called longitudinal waves.

Waves on springs or sound waves in air are examples of longitudinal waves.

(*ii*) **Transverse Waves** If the particles of the medium vibrates perpendicular to the direction of propagation of wave, the wave is called transverse waves.

Waves on strings under tension, waves on the surface of water are examples of transverse waves.

Electromagnetic Waves

The waves which do not require any medium for their propagation i.e. which can propagate even through the vacuum are called electromagnetic waves. They propagate as transverse wave.

- Electromagnetic waves of wavelength range 10^{-3} m to 10^{-2} m are called **microwaves**.
- Microwaves are absorbed by water molecules and living tissue internal heating will damage or kill cells.
- Microwaves are used in communications, satellites, telephony, heating water and food.

➠ **Note** Cathode rays, canal rays, α-rays, β-rays are not electromagnetic waves. Light and heat are examples of electromagnetic waves.

Spectrum of Electromagnetic Waves

Electromagnetic Waves	*Discoverer*	*Wavelength range (in metre)*	*Frequency range*	*Uses*
γ-rays	Henry Becqueral	10^{-14} to 10^{-10}	10^{20} to 10^{18}	Medical tracers, killing cancerous cells, steralisation, imaging defects in metal.
X-rays	W Roenetgen	10^{-10} to 10^{-8}	10^{18} to 10^{16}	Imaging defects in bones, hidden devices.
Ultraviolet rays	Ritter	10^{-8} to 10^{-7}	10^{16} to 10^{14}	Making vitamin D, making ions.
Visible radiation	Newton	3.9×10^{-7} to 7.8×10^{-7}	10^{14} to 10^{12}	Photography
Infrared rays	Hershel	7.8×10^{-7} to 7.8×10^{-3}	10^{12} to 10^{10}	TV remote control
Short radio waves or Hertz hertzian waves	Heinrich	10^{-3} to 1	10^{10} to 10^{8}	Communication, radio, TV.
Long radio waves	Marcony	1 to 10^{4}	10^{8} to 10^{6}	Communication, radio, TV.

Important Terms Related to Waves

- **Amplitude** (A) Maximum displacement of a vibrating particle of medium from its mean position is called amplitude.
- **Wavelength** (λ) Wavelength is the distance between any two nearest particle of the medium, vibrating in the same phase.
- **Frequency** (f) Frequency of vibration of a particle is defined as the number of vibrations completed by particle in one second.

$$\text{Frequency} = \frac{1}{\text{Time period}}$$

- Velocity of wave (v) = Frequency (f) × Wavelength (λ)

Waves in Different Frequency Range

According to their frequency range, longitudinal mechanical waves are divided into the following categories:

- The longitudinal mechanical waves which lie in the range 20 Hz to 20000 Hz are called **audible** or **sound waves**. Human ears are sensitive to the waves in this frequency range.
- The longitudinal mechanical waves having frequencies less than 20 Hz are called **infrasonic**. These are produced by earthquakes, volcanic eruption, ocean waves and elephants and whales.
- The longitudinal mechanical waves having frequencies greater than 20000 Hz are called **ultrasonic waves**.
- Human ear cannot detect the ultrasonic waves. But certain creatures like dog, cat, bat, mosquito can detect these waves. Bat also produces ultrasonic waves.
- Ultrasonic waves are used for sending signals, measuring the depth of sea, cleaning clothes and machinery parts of clocks removing lamp shoot from chimney of factories and in ultrasonography.
- The use of ultrasonic waves to investigate the action of the heart is called **echocardiography.**

Sound Waves

Sound wave is the form of energy which make us hear and having frequency and high wavelength. These can not travel in vacuum. It is of two types i.e. longitudinal wave and mechanical wave.

- The rebouncing back of sound when it strikes a hard surface is called **reflection of sound**. The reflection of sound is utilised in working of devices such as megaphone, bulbhorn, stethoscope and sound board.
- The repetition of sound caused by the reflection of sound waves is called an **echo**. In addition to curtains, carpets and sofa-sets in our rooms also reduce the formation of echoes by absorbing sound waves.
- The persistence of sound in a big hall due to repeated reflections from walls, ceiling and floor of the hall is **reverberation**.
- To hear echo, the minimum distance between the observer and reflection should be 17. Persistance of ear is 1/10 s.
- At the moon, the echo is not heared due to absence of atmosphere.

SONAR

It stands for sound navigation and ranging. It is used to measure the depth of a sea and to locate the submarines and shipwrecks. The transmitter of a sonar produces pulses of ultrasonic sound waves of frequency of about 50000 Hz. The reflected sound waves are received by the receiver.

Speed of Sound

The distance travelled by sound wave in one second is known as speed of sound wave. i.e.

Speed of sound, $v = \text{Frequency} \times \text{Wavelength}$

- Speed of sound basically depends upon elasticity and density of medium. The speed of sound in a medium is given by $v = \sqrt{\frac{E}{d}}$, where E is modulus of elasticity of the medium and d = density.
- For gases, $v = \sqrt{\frac{\gamma p}{d}}$

  where, p is pressure and γ is ratio of specific heats.
- Speed of sound is maximum in solids and minimum in gases. Speed of sound in air is 332 m/s, 1483 m/s in water and 5130 m/s in iron.
- When sound enters from one medium to another medium, its speed and wavelength changes but frequency remains unchanged.
- Speed of sound remains unchanged by the increase or decrease of pressure.
- The speed of sound increases with the increase of temperature of the medium. The speed of sound in air increases by 0.61 m/s, when the temperature is increased by 1°C.
- The speed of sound is more in humid air than in dry air because the density of humid air is less than the density of dry air.
- The speed of sound in air is very slower as compared to the speed of light in air. Therefore, in rainy season, the flash of lightning is seen first and the sound of thunder is heard a little later.

➠ **Note** The astronauts who land on moon, talk to one another through wireless sets using radio waves.

Characteristics of Sound Waves

There are following characteristics of sound waves.

Intensity

Intensity of sound at any point in space is defined as amount of energy passing normally per unit area held around that point per unit time. SI unit of intensity is W/m^2.

- Intensity of sound at a point is inversely proportional to the square of the distance of point from the source and directly proportional to the square of amplitude of vibration, square of frequency and density of the medium.
- The loudness depends on intensity as well as on sensitivity of ear.
- The loudness of sound is measured in decibel (dB).
- Sound level, $\beta = 10 \log_{10}\left(\frac{I}{I_0}\right)$

where, I_0 = the minimum intensity that can be heard called threshold of hearing 10^{-12} W/m^2 at 1 kHz.

Standard value of noise pollution

Areas	*Day time (dB)*	*Night time (dB)*
Industrial area	75	70
Commercial area	65	55
Residential area	55	45
Silence	50	40

➨ **Note** Likewise intensity of sound wave, intensity of an earthquake defines the energy released by it which is indicated by the local effects and potential for damage produced on the earth's surface. It can be measured by a seismograph.

Pitch

It is that characteristic of sound, which distinguishes a sharp sound from a grave sound.

- Pitch depends upon frequency of sound waves.
- The pitch of female voice is higher than the pitch of male voice.
- The pitch of sound produced by roaring of lion is lower whereas the pitch of sound produced by mosquito whisper is high.

Quality (or Timbre)

It is that characteristic of sound which enables us to distinguish between sounds produced by two sources having the same intensity and pitch.

SHOCK WAVES

A body moving with supersonic speed in air leaves behind it a conical region of disturbance which spreads continuously. Such a disturbance is called **shock waves.**

- These waves carry huge energy and may even make cracks in window panes or even damage a building.
- The speed of supersonic wave is measured in **mach number**. One mach number is the ratio of velocity of source to the velocity of sound.
- Mach number $= \frac{\text{Velocity of source}}{\text{Velocity of sound}}$
  - If mach number > 1, body is called supersonic.
  - If mach number > 5, body is called hypersonic.
  - If mach number < 1, body is said to be moving with subsonic speed.

Resonance

- If however, the frequency of the external force is equal to the natural frequency of the body, then the amplitude of the forced oscillations of the body becomes quite large. This phenomenon is called **resonance**.
- A group of soldiers on a bridge are advised not to walk in steps because their movement causes the bridge to vibrate. If they walk in step, the frequency of vibration may match the natural frequency of the bridge structure and thus causing resonance. This resonance of frequency can cause the bridge to collapse.

Some Definitions Regarding Waves

Interference When two waves of same frequency, same wavelength, same velocity moves in the same direction. Their superimposition results in the interference. In interference, energy is neither created nor destroyed but is redistributed.

Beat When two sound waves of slightly different frequencies, travelling in a medium along the same direction, superimpose on each other, the intensity of the resultant sound at a particular position rises and falls regularly with time. This phenomenon of regular variation in intensity of sound with time at a particular position is called beats.

Progressive Waves The disturbance produced in the medium travels onward, it being handed over from one particle to the next. Each particle executes the same type of vibration as the preceding one, though not at the same time.

Stationary Waves When combination of two waves moving in opposite directions, each having the same amplitude and frequency.

DOPPLER'S EFFECT

If there is a relative motion between source of sound and observer, the apparent frequency of sound heard by the observer is different from the actual frequency of sound emitted by the source. This phenomenon is called Doppler's effect.

When the distance between the source and observer decreases, the apparent frequency increases and *vice-versa*.

Uses of Doppler's Effect

(*i*) By police to check over speeding of vehicles.
(*ii*) At airport to guide the aircraft.
(*iii*) To study heart beats and blood flow in different parts on the body.
(*iv*) It is used to determine the velocities of which stars and galaxies are moving toward or away from the earth.

OPTICS

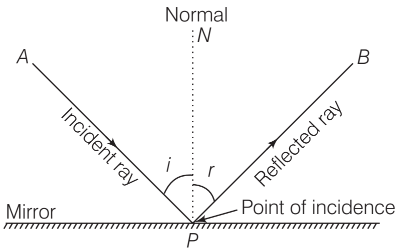
LIGHT

It is the radiation which makes our eyes able to see the object. Its speed is 3×10^8 m/s in vacuum.

- Light is the form of energy. It is a transverse wave and it takes 8 min 19 s to reach on the earth from the sun.
- The light reflected from the moon takes 1.28 s to reach the earth.
- It represents the phenomenon of reflection, refraction, interference, diffraction, scattering and polarisation.

Reflection of Light

Reflection is the phenomenon of change in the path of light without any change in its medium. In reflection, the frequency, speed and wavelength do not change, but a phase change may occur depending on the nature of reflecting surface.



Reflection of Light

Laws of Reflection

- The angle of reflection $(\angle r)$ is always equal to the angle of incidence $(\angle i)$ i.e. $\angle i = \angle r$.
- The incident ray, the reflected ray and the normal at the point of incidence all lie in the same plane.

MIRROR

It is a polished surface (one side) like glass, which reflects almost all the light that is incident on it.

There are two types of mirror

Plane Mirror

If the reflecting surface of a mirror is plane, then the mirror is called a **plane mirror**.

- Size of image is always equal to the size of object.
- The minimum size of the mirror required to see the full image of an observer is half the height of the observer.
- If the plane mirror is rotated in the plane of incidence by an angle θ, then the reflected ray rotates by an angle 2θ.
- If the object is displaced by a distance x towards or away from the mirror, then its image will be displaced by a distance x towards or away from the mirror.
- Focal length of plane mirror is infinity i.e. power of the plane mirror is zero.
- Linear magnification produced by plane mirror is 1.
- When two plane mirrors are kept facing each other at an angle θ and an object is placed between them, then

  (*i*) Number of images, $n = \dfrac{360°}{\theta} - 1$

  If $\dfrac{360°}{\theta}$, is even or object lies symmetrically.

  (*ii*) Number of images, $n = \left(\dfrac{360°}{\theta}\right)$. If $\left(\dfrac{360°}{\theta}\right)$ is odd or the object lies symmetrically.

Uses of the Plane Mirror

(*i*) Plane mirrors are used as looking glass.

(*ii*) Plane mirrors are used in constructing periscope which is used in submarines.

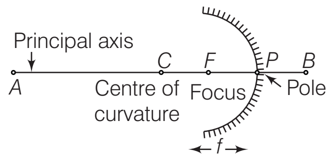
(*iii*) Plane mirror are used to make kaleidoscope.

Spherical Mirrors

Spherical mirrors can be regarded as a part of a hollow reflecting spheres of glass. These are of two types

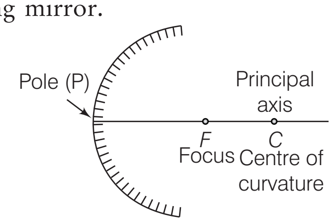
(*i*) Concave mirror (*ii*) Convex mirror

Concave mirror The spherical mirror whose reflecting surface is inwardly leaned, called concave mirror. It is also called converging mirror.



Concave mirror

Convex mirror The spherical mirror whose reflecting surface is outwardly leaned, called convex mirror. It is also called diverging mirror.



Convex mirror

Real Image

- Real images can be formed on a screen by the actual intersection of reflected (or refracted) light rays.
- The image formed on a cinema screen is a real image.
- A concave mirror forms real image, when an object is placed beyond its focus.

Virtual Image

- Virtual images cannot be formed on the screen by the apparent intersection of reflected (or refracted) light rays.
- The image of our face when we look into a plane mirror is a virtual image. A concave mirror forms a virtual image, when an object is placed between focus and pole of the mirror.

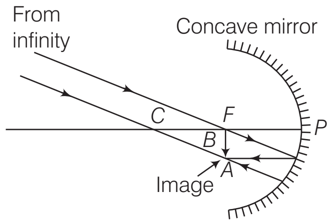
Position of object	*Position of image*	*Size of image*	*Nature of image*
At infinity	At *F*	Highly diminished	Real and inverted
Between infinity and *C*	Between *F* and *C*	Diminished	-do-
At *C*	At *C*	Same size	-do-
Between *F* and *C*	Between infinity and *C*	Enlarged	-do-
At *F*	At infinity	Highly enlarged	-do-
Between *F* and *P*	Behind the pole	Enlarged	Virtual and erect

Position of object	*Position of image*	*Size of image*	*Nature of image*
At infinity	At *F*	Highly diminished	Erect and virtual
Between infinity and pole	Between *F* and *P*	Diminished in size	Erect and virtual

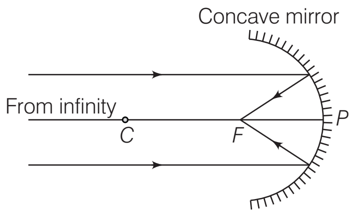
Image Formation by Concave Mirror

The following are the positions and nature of the images formed in a concave mirror for different positions of the object

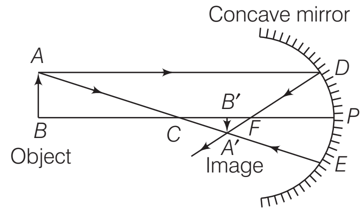
(*i*) **When the object is at infinity** Image is formed at *F*.



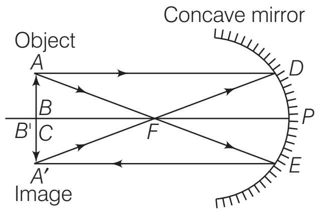
(*ii*) **When the rays are parallel to the principal axis** The image is formed at *F*.



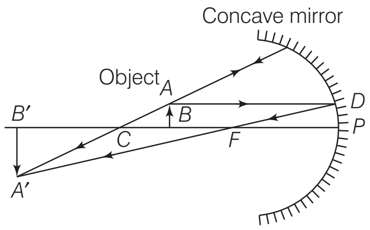
(*iii*) **When the object is between infinity and *C*** The image formed is real, inverted and smaller in size and is formed between *F* and *C*.



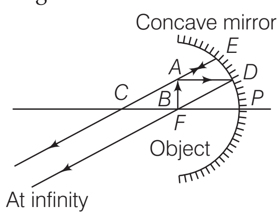
(*iv*) **When the object is at *C*** The image formed is inverted, real, equal in size and is formed at *C* itself.



(*v*) **When the object is between *F* and *C*** The image is formed beyond *C* and is inverted, real and larger than the object.



(*vi*) **When the object is at focus** The image is formed at infinity. The image is real, inverted and infinitely large in size.



(*vii*) **When the object is between *F* and *P*** The images formed behind the pole. The image is virtual, erect and large.